

# WEEK 7 – Final Risk Modeling & Output Generation

(alphapulse\_final\_output Implementation)

## Objective

To finalize financial risk computations and generate structured outputs for visualization and reporting.

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## 1. Data Preparation & Cleaning

In Week 7, the cleaned dataset (week5\_finance\_cleaned.csv) was used as input.

### Steps Performed:

- Loaded dataset using Pandas
- Converted date column to datetime
- Sorted data chronologically
- Verified missing/null values
- Ensured adjusted close price usage (for split/dividend correction)

### Why This Is Important:

Financial modeling requires:

- Clean chronological data
  - No missing return values
  - Corrected stock price adjustments
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## 2. Daily Log Return Calculation

### Formula Used:

$$R_t = \ln\left(\frac{P_t}{P_{t-1}}\right)$$

Where:

- $P_t$  = Current closing price
- $P_{t-1}$  = Previous day closing price

### Implementation:

Used NumPy vectorized operations for fast computation.

### Why Log Returns?

- Additive over time
  - Better statistical properties
  - Preferred in quantitative finance
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### 3. Portfolio Return Calculation

Since multiple stocks were selected:

- Equal weight assumption applied
- Mean return calculated across assets

#### Formula:

Portfolio Return =  $\sum w_i R_i$

Where:

- $w_i$  = weight of stock
- $R_i$  = daily return

This created:

Portfolio\_Return

This became the base for all further risk analysis.

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### 4. Rolling Volatility (30-Day)

#### Formula:

Rolling Volatility =  $\text{Std}(\text{Portfolio Return over 30 days})$

#### Why 30 Days?

- Represents 1 trading month
- Standard industry practice

#### Output:

Created:

rolling\_volatility\_30

Exported as:

rolling\_volatility.csv

This measures market uncertainty.

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## 5. Monte Carlo Simulation (10,000 Runs)

### Purpose:

To simulate possible future portfolio performance over 1 year.

### Steps:

1. Calculated:
  - Historical mean return
  - Historical standard deviation
2. Generated random returns:  
Using normal distribution:

$N(\mu, \sigma)$

3. Simulated 252 trading days
4. Repeated 10,000 times

### Why 10,000 Runs?

- Ensures statistical stability
  - Industry-level robustness
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## 6. Value at Risk (VaR – 95%)

### Definition:

Maximum expected loss with 95% confidence.

### Formula:

$VaR_{95} = 5th \text{ percentile of simulated returns}$   
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If result is:

-0.0369

That means:

You could lose **3.69%** with 95% confidence.

**Output:**

Exported as:

var\_results.csv

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**7. Final Outputs Generated (alphapulse\_final\_output)**

week5\_finance\_cleaned.csv

rolling\_volatility.csv

var\_results.csv

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**8. Conclusion**

These files were structured specifically for Tableau ingestion.